

RETRO HORROR CAMERA KIT

PS1 / PSX and VHS / CRT camera effects

for Unity 6 · Universal Render Pipeline · Render Graph

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BY MIDNIGHT RENTAL
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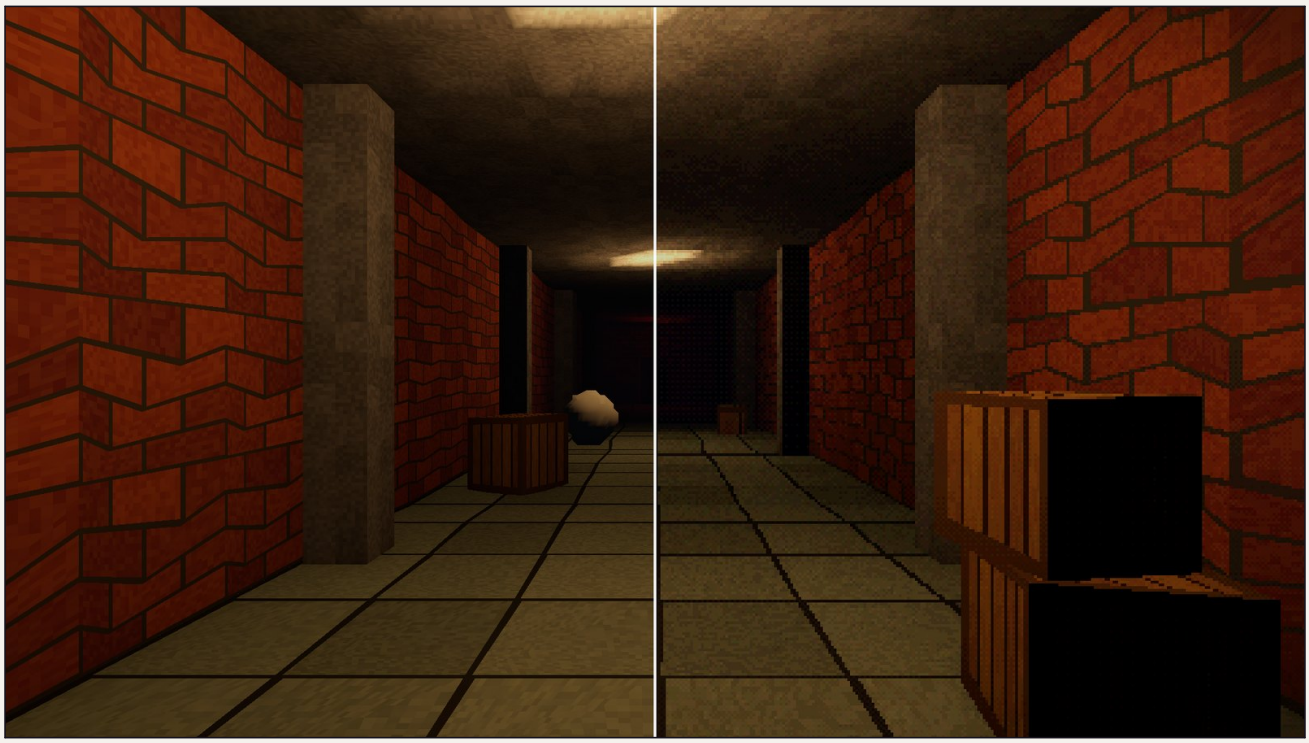
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01 WELCOME

Retro Horror Camera Kit turns a Unity 6 URP camera into a PS1 survival horror camera, a found-footage camcorder or a late-night broadcast on an old CRT. It is written from scratch for **Render Graph** and configured entirely through URP's **Volume** system, so it behaves like URP's own post-processing.



Left: URP as is. Right: the PS1 Survival Horror look.

What is inside

Part	What it does	Set up with
PSX full-screen effect	Low resolution, reduced color depth, ordered dither.	Volume component
VHS / CRT full-screen effect	Tape noise, tracking errors, color bleed, scanlines, CRT curvature, camcorder overlay.	Volume component
PSX Lit material	Vertex jitter, affine texture warping and vertex lighting on your meshes.	Material
Retro Sky and **Neon Grid**	Procedural retro skyboxes and a glowing grid floor.	Material
Presets	16 looks, 6 skies and a neon grid, ready to apply.	Setup Window
Setup Window	Configures every URP renderer in the project and applies presets.	Tools menu

02 GETTING STARTED

Requirements

Unity	6.0 LTS (6000.0) or newer. Tested on 6.0 and 6.3.
Render pipeline	Universal Render Pipeline 17.0 or newer
Rendering path	Forward or Forward+ (Deferred with limitations, see Known Limitations)
Render Graph	Required. URP Compatibility Mode is not supported.
Dependencies	None. Only URP and Unity's core render pipeline library.

Quick start

1 Open the Setup Window

Use `Tools > Retro Horror Kit > Setup Window`.

2 Add the renderer features

Under **1. Renderer Setup**, click **Add Missing Renderer Features**. This adds the PSX and VHS features to *every* URP renderer your project uses, including those of other quality levels (for example PC and Mobile).

3 Pick a look

Under **2. Look**, click **Apply** next to a preset such as *PS1 Survival Horror* or *Found Footage*. The kit creates a global Volume called `Retro Horror Kit Volume`.

4 Optional: pick a sky

Under **3. Sky**, apply a sky. The scene fog it was designed for is applied with it.

> **TIP** Make sure your camera has **Post Processing** enabled (Camera > Rendering > Post Processing). The effects follow this checkbox per camera.

> **NOTE** To customize a preset, duplicate it first (Ctrl+D in the Project window) and assign the copy to the Volume. Editing a preset directly changes it everywhere it is used.



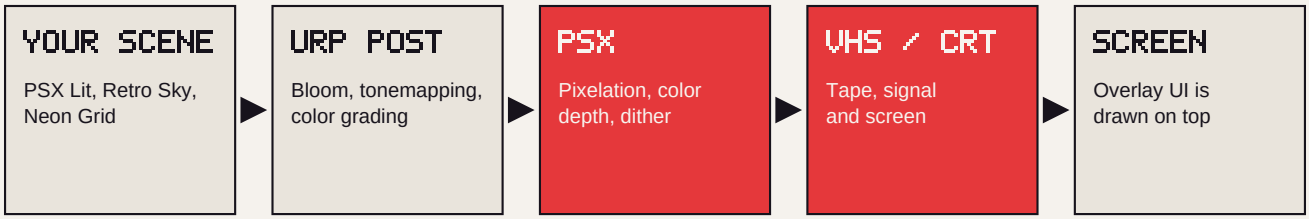
PART 1

FULL-SCREEN EFFECTS

PSX and VHS / CRT, driven by URP Volumes

03 HOW IT WORKS

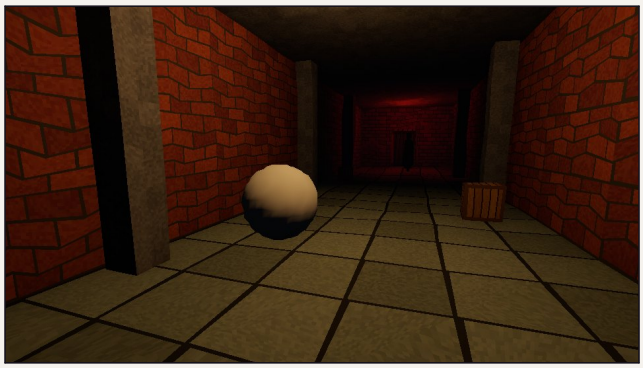
The full-screen effects are renderer features configured with Volume components, exactly like URP's own effects: add **Retro Horror Kit/PSX** and **Retro Horror Kit/VHS** to any Volume Profile, use local Volumes to change the look per area and blend between them. A setting that is not overridden does nothing.



Topic	Behavior
Order	Both effects run after URP's post-processing. PSX always runs before VHS, regardless of the order of the features in the renderer: the image is pixelated first, then "recorded to tape and played on a CRT".
Cameras	Every camera with Post Processing enabled, including the Scene view. Preview and reflection cameras are skipped.
UI	Screen Space - Overlay canvases are drawn after the effects and stay clean. Screen Space - Camera and World Space UI go through the effects.
Cost when off	Disabled settings are compiled out or skipped. If a whole component is inactive, its render pass is not added at all.

04 PSX VOLUME

Add with `Add Override > Retro Horror Kit > PSX`.



Without the effect

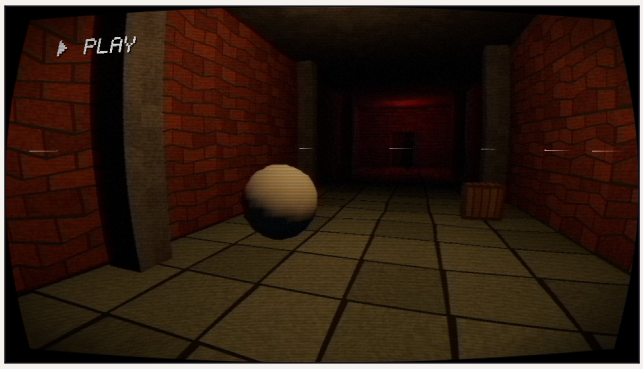


160p, 12 color levels, full dither

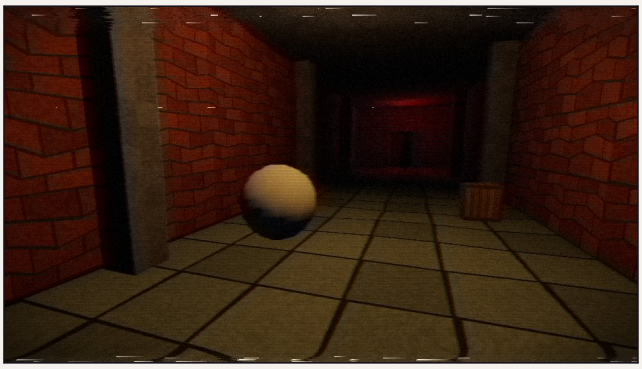
SETTING	DESCRIPTION
Pixelation Mode	*Off*, *Pixel Size* or *Target Height*.
Pixel Size	*Pixel Size* mode: each low-res pixel covers N x N screen pixels.
Target Height	*Target Height* mode: the vertical resolution to aim for. Most PS1 games ran at 240. The kit picks the **integer** pixel size closest to it, so every pixel has the same size; the result is approximate (270 lines at 1080p for a target of 240).
Color Levels	Levels per color channel. 256 = off. The PS1 frame buffer used 32 (15-bit color). Reduction happens in display (sRGB) space, so steps stay even in dark areas.
Dither Strength	4x4 Bayer ordered dither before color reduction, the pattern the PS1 GPU used. 0 = off. With pixelation on, the pattern follows the large pixels. Even at 256 levels a little dither smooths gradient banding.

05 VHS VOLUME

Add with `Add Override > Retro Horror Kit > VHS`. Every effect is off (0) by default.



Late Night Channel: curvature, scanlines, PLAY label



Cursed Rental: heavy chroma bleed and tracking errors

SETTING	DESCRIPTION
Scanline Intensity	Darkness of the scanlines.
Scanline Count	Lines across the screen height, independent of resolution. Values close to the real screen height can cause moire.
Chromatic Offset	Horizontal red / blue separation in screen pixels.
Chroma Bleed	Smears color to the right while brightness stays sharp, like the low color resolution of tape (NTSC YIQ). The most "VHS" setting of all.
Grain Intensity	Per-pixel tape grain. Changes 60 times per second, independent of frame rate.
Band Noise	Short bright specks on random tape lines.
Tracking Strength	Line jitter, a rolling band of strong distortion and head-switching noise along the bottom edge.
Tracking Speed	How fast the band rolls down the screen.
Curvature	Barrel distortion of the CRT glass, with anti-aliased black borders.
Vignette Intensity / Smoothness	Darkening towards the corners.

> **GOOD TO KNOW** The VHS effects keep animating while the game is paused (Time.timeScale = 0), like a tape that keeps playing behind a pause menu.

06 CAMCORDER OVERLAY

Part of the VHS Volume. A built-in bitmap font draws it, so no font asset is needed. The text is burned into the tape: it bends with the curvature, jitters with tracking errors and gets scanlines and noise like the picture.



Found Footage 1998: blinking REC, custom clock.

SETTING	DESCRIPTION
Overlay	Enables the overlay.
Overlay Label	*None*, *Rec* (blinking dot), *Play* or *Pause*, top-left.
Overlay Timestamp	Time and date, bottom-left, e.g. PM 11:47:05 / OCT. 31 1998.
Timestamp Source	*System Clock*: the player's local time. *Custom*: starts at the date and time below and runs from application start.
Custom Year ... Minute	Start of the custom clock. Invalid days (31 Feb) are clamped.
Overlay Size	Text height as a fraction of the screen height.
Overlay / Record Dot Color	Text color and the color of the REC dot.



PART 2

MATERIALS & SKIES

PSX Lit, Retro Sky and Neon Grid

07 PSX LIT MATERIAL

Shader `Retro Horror Kit/PSX Lit`. Use it on your meshes for the parts of the PS1 look a post-process cannot do. Its inspector groups settings into Surface, Lighting, Emission, PSX and Advanced.

Lighting modes

Mode	Description
Vertex	Gouraud shading: lighting per vertex, like the PS1. Works with Forward and Forward+. Shadows come from the main (directional) light only. Needs reasonably dense geometry: a large quad with four vertices gets almost no light from a point light in its middle.
Pixel	Smooth per-pixel diffuse lighting with shadows from all lights. Use it for flashlights and lamps that should cast shadows.
Unlit	Texture and color only. Fog still applies.

All modes are diffuse only (no specular, like the PS1) and receive URP fog and screen-space ambient occlusion. Alpha clipping sets the render queue automatically.

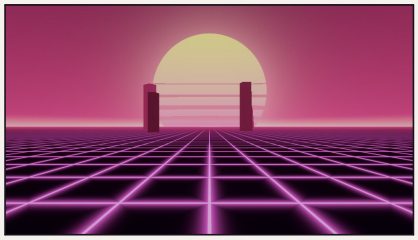
PSX settings

SETTING	DESCRIPTION
Vertex Snapping	Snaps vertices to a low-resolution screen grid: geometry wobbles as the camera moves.
Snap Resolution	Vertical resolution of that grid. Lower = stronger wobble. Matching the PSX Volume's target height looks consistent.
Affine Texture Mapping	Textures are interpolated without perspective correction and warp across large polygons.
Affine Strength	Blend between affine (1) and correct (0) mapping. Lower it if large floors warp too much; PS1 games subdivided large surfaces for the same reason.

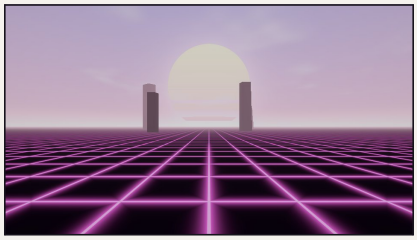
> **TIP** For an authentic look use small textures (64-256 px) with **Filter Mode: Point** and **Generate Mip Maps** off, subdivide large surfaces into roughly 1 m cells and keep polygon counts low.

08 RETRO SKY

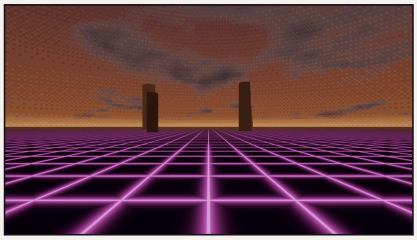
Shader `Retro Horror Kit/Retro Sky`. Fully procedural: no textures, no seams, sharp at any resolution. Every layer can be turned off, and disabled layers are not compiled into the material. Use **Apply to Scene** at the top of its inspector, or set it under Window > Rendering > Lighting > Environment.



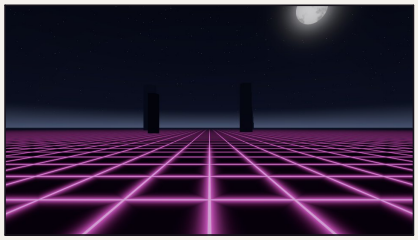
Synthwave Sunset
Striped sun, pink horizon, faint stars.



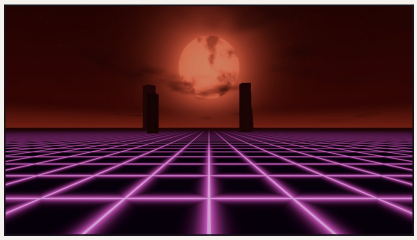
Pastel Vaporwave
Soft pastels and drifting clouds.



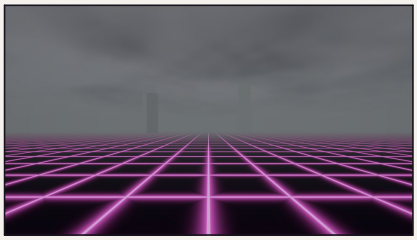
PS1 Dusk
Low-res, dithered sunset, as on the console.



Night Stars
Cratered moon and twinkling stars.



Blood Moon
A red moon behind dark clouds.



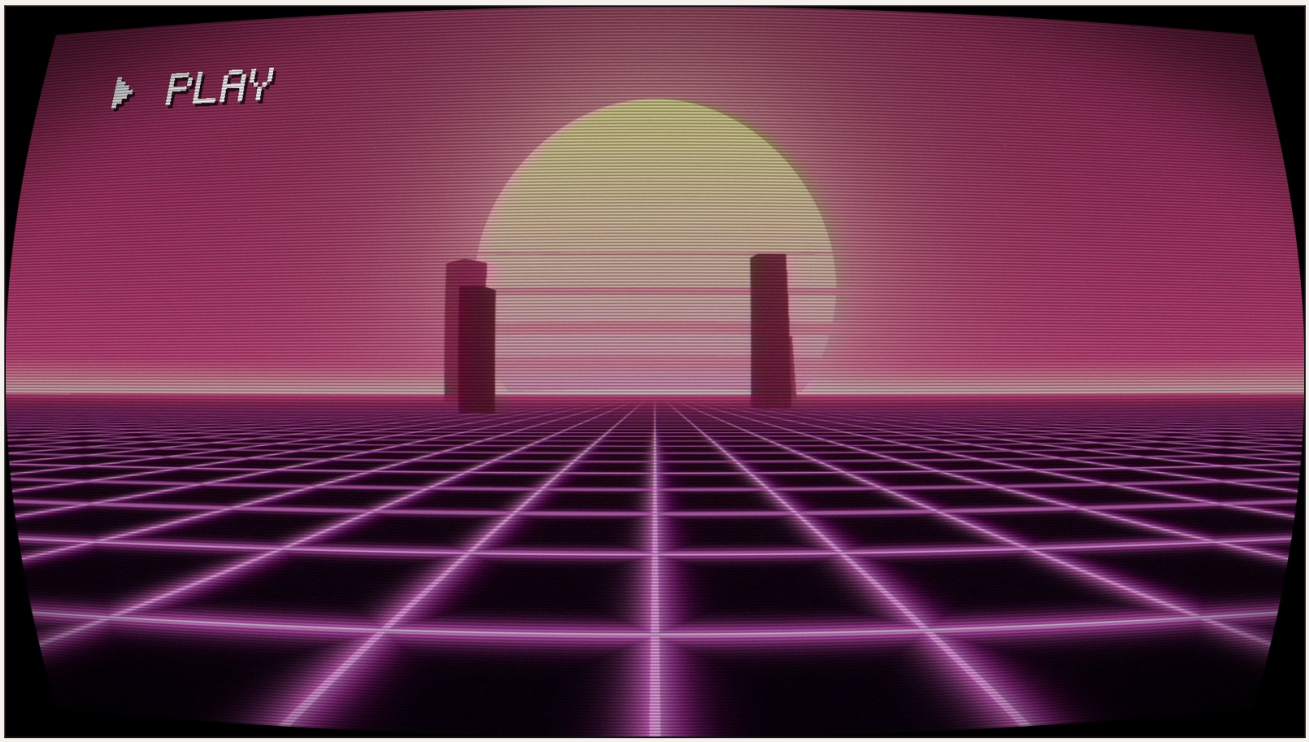
Overcast Fog
Grey cloud cover fading into fog.

Layers

SETTING	DESCRIPTION
Gradient	Zenith, horizon and ground colors, falloffs and a horizon glow band.
Sun	Synthwave sun with a top-to-bottom gradient and scrolling stripes that thicken towards the bottom. Sets behind the horizon. Positioned by azimuth / elevation.
Moon	Procedural craters, darkened limb and glow. HDR color for a blood moon with bloom.
Stars	Individual brightness and twinkle, fading towards the horizon.
Clouds	A cloud layer with perspective towards the horizon, two colors and wind.
PS1 Look	Snaps the sky to a coarse grid like a low-res skybox texture, with reduced colors and dither. The pattern is fixed to the sky, not the screen.
Horizon Fog	Blends the sky into the scene fog color near the horizon: no hard sky line in foggy scenes.
Suggested Scene Fog	The fog the sky is designed for. Applied together with the sky by Apply to Scene, the Setup Window and RetroSkyUtility. **Copy From Current Scene Fog** stores your own.

09 NEON GRID

Shader `Retro Horror Kit/Neon Grid`. An unlit glowing grid floor. Lines are anti-aliased and turn into an even tint in the distance instead of shimmering.



Neon Grid under Synthwave Sunset, Late Night Channel look.

SETTING	DESCRIPTION
Base Color / Line Color	Floor and line colors. Line Color is HDR: values above 1 glow with Bloom.
Grid Space	*World*: cells are always Cell Size units and continue across separate meshes. *UV*: follows the mesh UVs, for walls and ramps.
Cell Size / Line Width	Cell size in world units; line width as a fraction of a cell.
Glow Width / Intensity	Soft glow around the lines.
Scroll Speed	Moves the grid, for the classic "driving into the horizon" shot.
Fade Start / End	Lines fade out with distance from the camera.

TIP For a seamless horizon use scene fog in the horizon color. The synthwave skies suggest one.



PART 3

PRESETS & DEMOS

16 looks, 6 skies, 2 demo scenes

10 LOOK PRESETS

Volume Profiles in `Presets/Volumes`. All shown from the same camera in the demo corridor. Some also use URP's own Bloom or Color Adjustments.



PS1 Survival Horror
240p, 15-bit color, dither, soft vignette. The classic.



PS1 Clean
240p and 15-bit color without dither.



PS1 Nightmare
160p, heavy color reduction, grain.



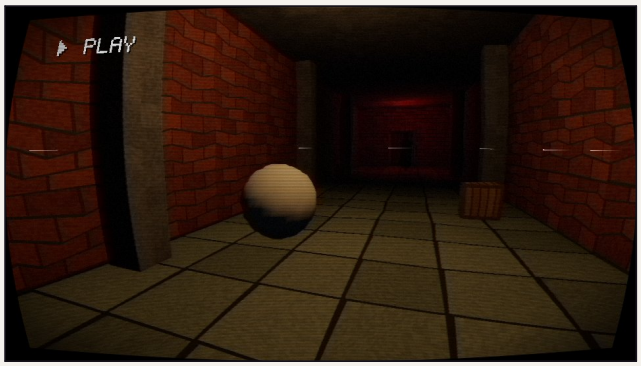
Pixel Purist
Pixelation only.



Found Footage
Camcorder tape with REC and the player's real date and time.



Found Footage 1998
Same, with a clock starting on Oct. 31 1998, 23:47.



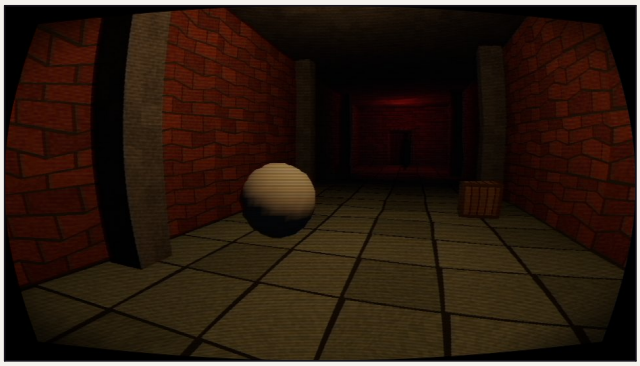
Late Night Channel
80s broadcast on a curved CRT, PLAY label, bloom.



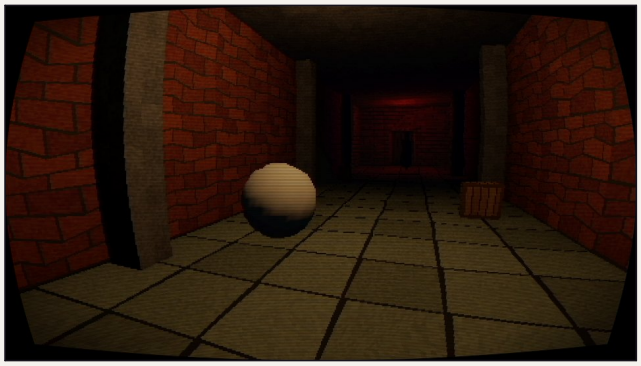
Haunted Tape
Damaged tape: tracking errors, noise and bleed, PAUSE label.



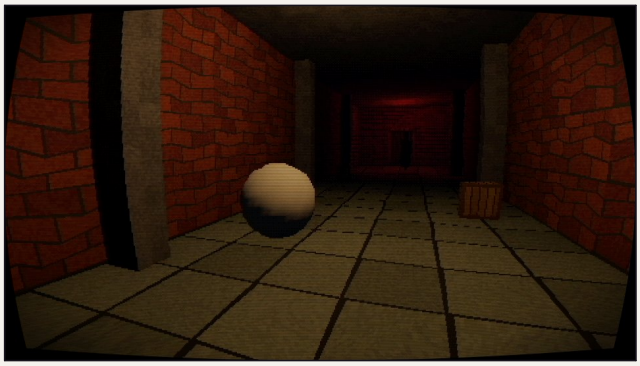
CCTV
Black-and-white security camera with timestamp.



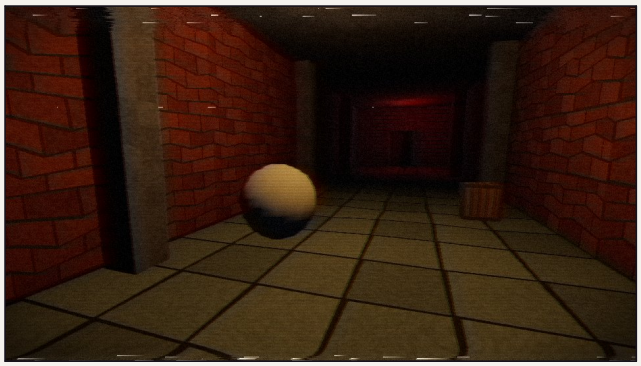
Old CRT Monitor
Strong curvature, scanlines and vignette.



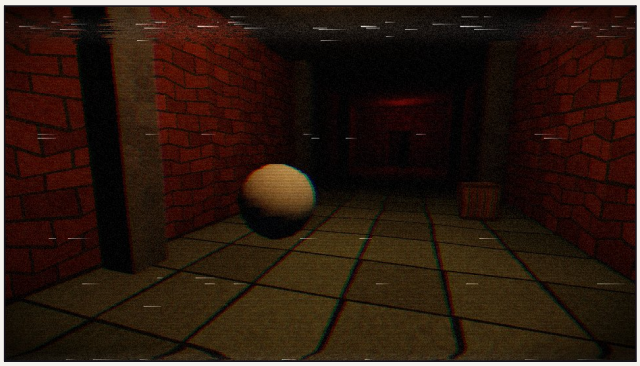
Arcade Cabinet
240p on a curved arcade screen with bloom.



Console On TV
A PS1 game on a living-room CRT.



Cursed Rental
Worn rental tape with heavy color bleed.



Nightmare Broadcast
A signal that is barely holding together.



Liminal Dream
Soft, bloomy, slightly dithered dream look.



Subtle Retro
A light touch for a hint of retro.

11 DEMO SCENES

`Demo/Scenes/Demo_Corridor` (indoor horror) and `Demo/Scenes/Demo_Synthwave` (outdoor, skies and neon grid). Run the setup (Getting Started, step 2) first, then press Play.

INPUT	ACTION
Mouse	Hold to look around
W A S D	Move
Q E	Down / up
SHIFT	Move faster
◀ ▶	Previous / next look preset
▲ ▼	Next / previous sky (synthwave demo)
P	Effects on / off
H	Hide / show the help text

The demo scripts use the Input System package when it is installed and enabled, and the legacy Input Manager otherwise. The **Demo** folder can be deleted without affecting the kit.



The corridor demo running as a Windows build, with its on-screen help.

12 SCRIPTING

All runtime types are in the `RetroHorrorKit` namespace.

Switch the whole look

```
using UnityEngine;
using UnityEngine.Rendering;

public class LookSwitcher : MonoBehaviour
{
    [SerializeField] Volume volume;
    [SerializeField] VolumeProfile tapeLook;

    public void UseTapeLook() => volume.sharedProfile = tapeLook;
}
```

Drive a single setting

For example more tracking noise as the player loses their sanity:

```
using RetroHorrorKit;
using UnityEngine;
using UnityEngine.Rendering;

public class SanityEffect : MonoBehaviour
{
    [SerializeField] Volume volume;
    [Range(0f, 1f)] public float sanity = 1f;

    VHSVolume m_Vhs;

    void Start()
    {
        // volume.profile returns a per-instance copy,
        // so the preset asset is not modified.
        volume.profile.TryGet(out m_Vhs);
    }

    void Update()
    {
        if (m_Vhs == null)
            return;
        m_Vhs.trackingStrength.Override(1f - sanity);
        m_Vhs.bandNoise.Override((1f - sanity) * 0.8f);
    }
}
```

Change the sky with its fog

```
RetroHorrorKit.RetroSkyUtility.ApplyToScene(nightSkyMaterial);
```

13 PERFORMANCE

Feature	Cost
PSX, pixelation on	Two full-screen passes, the first at low resolution.
PSX, color only	One full-screen pass.
VHS	One full-screen pass. Chroma Bleed adds 6 texture reads per pixel, Chromatic Offset 2. The overlay costs only inside the text area.
PSX Lit, Vertex lighting	Per vertex x visible lights. Cheap for low-poly scenes. With Forward+, every visible light is evaluated per vertex.
PSX Lit, Pixel lighting	Comparable to URP Simple Lit.
Retro Sky	Only enabled layers are compiled. Clouds are the most expensive layer (4 noise octaves).

Inactive settings and components cost nothing. The VHS shader uses uniform branches instead of shader variants, so turning its effects on and off at runtime never causes a shader compile hitch.

14 TROUBLESHOOTING

I don't see any effect.

Open the Setup Window: every renderer should show PSX and VHS as set up. The camera needs **Post Processing** enabled. The Volume must be global (or the camera inside a local Volume) with a weight above 0. The kit's Volume has priority 10; a Volume with a higher priority overriding the same settings wins. The Setup Window shows an error if URP Compatibility Mode is enabled.

It works on PC but not on mobile (or another quality level).

Each quality level can use its own URP asset and renderer. **Add Missing Renderer Features** covers all of them.

It works in the editor but not in a build.

The renderer features keep a reference to their shader. If you added a feature by hand and its Shader field is empty, assign Hidden/RetroHorrorKit/PSX or Hidden/RetroHorrorKit/VHS, or remove and re-add it.

The sky has a hard line at the horizon in a foggy scene.

Raise *Horizon Fog > Blend To Fog Color* on the sky, or use **Apply to Scene** to set the sky's suggested fog.

My PSX Lit floor is dark under a point light.

In Vertex mode, light is evaluated only at vertices. Subdivide the floor (about 1 m cells) or switch the material to Pixel lighting.

The custom overlay clock resets in the editor.

It counts from application start; in the editor this restarts on every domain reload, including entering Play Mode.

15 KNOWN LIMITATIONS

Area	Limitation
Render Graph	URP Compatibility Mode is not supported.
PSX Lit	No baked lightmaps (ambient light and Light Probes work). Not GPU Resident Drawer compatible; such objects fall back to regular rendering. In Vertex mode only the main light casts shadows on it.
Deferred	PSX Lit renders forward-only; SSAO on PSX Lit objects is not supported in Deferred.
Unity 6.0 + Render Texture	With Render Scale below 1, URP 6.0 crops the image of a camera that renders into a Render Texture when any effect runs after post-processing (URP bug, fixed in 6.1+). Screen cameras are not affected. On 6.0, keep Render Scale at 1 for such cameras.
Not yet tested	XR / VR, Gamma color space, consoles and WebGL. The code accounts for XR and Gamma, but they have not been verified. Tested on Windows with DirectX 12: Unity 6.0 and 6.3 in the editor, 6.3 in a player build.

A MIDNIGHT RENTAL PRODUCTION

THANKS FOR WATCHING

■ STOP

Questions, bugs or ideas? Reach out through the store page you bought the kit from.

If the kit helped your game, a short review means a lot.